

Foundations & Linear Regression

INFO 521 · Module 1

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Learning outcomes

By the end of this module you should be able to:

1. **Formulate the linear regression problem** — design matrix, weights, loss.
2. **Solve for the least-squares estimate** — normal equations and geometric interpretation.
3. **Interpret residuals and fitted values** — diagnostics and model assessment.

The regression problem

Given N observations $(\mathbf{x}_n, y_n)$:

- **Design matrix** $\mathbf{X} \in \mathbb{R}^{N \times D}$: one row per observation, one column per feature.
- **Target vector** $\mathbf{y} \in \mathbb{R}^N$: continuous outcomes.
- **Weights** $\mathbf{w} \in \mathbb{R}^D$: the parameters we solve for.

Prediction: $\hat{y}_n = \mathbf{w}^\top \mathbf{x}_n$

Loss: Mean squared error

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (y_n - \mathbf{w}^\top \mathbf{x}_n)^2$$

The normal equations

Minimizing the loss gives:

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

This is the **closed-form solution** — no iteration needed. The $(\mathbf{X}^\top \mathbf{X})^{-1}$ term assumes the design matrix is full rank.

Interpretations: - $\mathbf{X}^\top \mathbf{X}$ is proportional to the *Gram matrix* (feature covariances). - $\mathbf{X}^\top \mathbf{y}$ is the *cross-covariance* between features and targets. - $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is the *pseudo-inverse* of $\mathbf{X}$.

Try it: Interactive tool

Take 5 minutes to explore the least-squares tool above. Adjust the data and weights, observe the loss change, and then complete the mastery quiz ($\geq 80\%$ reveals your token).